PRACTICAL LAB: JAVASCRIPT OPERATORS MASTERY

Course: DFK20013 Web Design Technology

Topic: 3.1.1 Variables, Data Types, and Operators

Duration: 60 – 90 Minutes

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Date	**2/3/2026**

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1. Learning Outcomes

By the end of this lab, students should be able to:

- Implement **Arithmetic Operators** for calculations.
- Use **Assignment Operators** to update values efficiently.
- Control application states using **Increment/Decrement**.
- Validate user access using **Relational** and **Logical** operators.

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2. Practical Task: The All-in-One Operator Workshop

Create a file named lab_operators.html. Copy and complete the coding below.

HTML

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <title>DFK20013: JS Operators Workshop</title>

    <style>

        body { font-family: 'Segoe UI', sans-serif; max-width: 850px;
margin: 2rem auto; background-color: #f0f2f5; padding: 20px; color: #333;
}

        section { background: white; padding: 25px; border-radius: 12px;
margin-bottom: 25px; box-shadow: 0 4px 6px rgba(0,0,0,0.1); }

        h2 { color: #2c3e50; border-left: 5px solid #3498db; paddingleft:
10px; margin-top: 0; }

        .output-box { background: #2c3e50; color: #27ae60; padding: 15px;
border-radius: 6px; font-family: 'Courier New', monospace; margin-top:
10px; }
```

```
        .short-note { background: #fff3cd; color: #856404; padding: 15px;
border-radius: 6px; margin-top: 15px; border-left: 5px solid #ffc107;
display: none; font-size: 0.9rem; }

        input { padding: 8px; width: 100px; border-radius: 4px; border:
1px solid #ccc; }

        button { padding: 8px 20px; background: #3498db; color: white;
border: none; border-radius: 4px; cursor: pointer; font-weight: bold; }
button:hover { background: #2980b9; }

        code { background: #eee; padding: 2px 4px; border-radius: 3px;
font-weight: bold; color: #d63384; }

    </style>

</head>
```

```
<body>



    <header style="text-align: center; margin-bottom: 30px;">

        <h1>JavaScript Operators Mastery</h1>

        <p>Follow the steps and observe the Console and UI outputs.</p>

    </header>
```

```
    <section>

        <h2>Task 1: Basic Arithmetic (Check Console)</h2>

        <p>This task demonstrates background calculations using
<code>+</code>, <code>-</code>, <code>*</code>, <code>/</code>,
<code>**</code>, and <code>%</code>.</p>

        <button onclick="runTask1()">Run Console Logic</button>

    </section>
```

```
    <section>

        <h2>Task 2: POS System (Arithmetic & Assignment)</h2>

        <p>Price: RM 50.00. <strong>Promo:</strong> Buy > 4 units to get
RM 20.00 discount.</p>

        Quantity: <input type="number" id="qty" value="1">

        <button onclick="runTask2()">Calculate Total</button>

        <div id="outPOS" class="output-box">Final Total: RM 0.00</div>

        <div id="notePOS" class="short-note"></div>

    </section>
```

```
    <section>
        <h2>Task 3: Stock Control (Increment & Decrement)</h2>
        <p>Simulating inventory updates.</p>
        Stock Level: <span id="stockVal" style="font-size: 1.5rem;
fontweight: bold;">10</span><br><br>
        <button onclick="runTask3('inc')">Add (+)</button>
        <button onclick="runTask3('dec')"
style="background:#6c757d;">Remove (-)</button>
        <div id="noteStock" class="short-note"></div>
    </section>
```

```
    <section>
        <h2>Task 4: Gate Access (Relational & Logical)</h2>
        <p>Rules: Age 18+ OR (Age 13+ AND Parent present).</p>
        Age: <input type="number" id="age" placeholder="Age">
        <label><input type="checkbox" id="parent"> With
Parent?</label><br><br>
        <button  onclick="runTask4()"  style="background:#27ae60;">Verify
Entry</button>
        <div id="outSecurity" class="output-box">Status: Waiting...</div>
        <div id="noteSecurity" class="short-note"></div>
    </section>
```

Inside the <script> tag, type all the following code:

JAVASCRIPT:

1.

```
<script>
    // Helper to show explanations
    function showNote(id, text) {
        const noteDiv = document.getElementById(id);
        noteDiv.style.display = 'block';
        noteDiv.innerHTML = text;
    }
```

2.

```
// --- TASK 1: ARITHMETIC (BASIC) ---
function runTask1() {
    console.log("--- Task 1: Arithmetic Operations ---");
    let students = 31;
    students = students + 1;   // Addition
    students = students - 2;   // Subtraction
    students = students * 2;   // Multiplication
    students = students / 2;   // Division
    let power = 10 ** 2;       // Exponentiation (100)
    let remainder = 31 % 2;    // Modulus (1 - Odd)

    console.log("Final Student Count: " + students);
    console.log("10 to the power of 2: " + power);
    console.log("Remainder of 31/2: " + remainder);
    alert("Check Browser Console (F12) for results!");
}
```

3.

```
// --- TASK 2: POS SYSTEM (ARITHMETIC & ASSIGNMENT) ---
function runTask2() {
    let price = 50;
    let qty = parseInt(document.getElementById('qty').value);

    // a. Arithmetic Operator (*)
    let total = price * qty;
    let note = "<b>Subtopic (a) Arithmetic:</b> <code>*</
    code> multiplied " + price + " by " + qty + ".<br>";

    // b. Assignment Operator (-=)
    if (qty > 4) {
        total -= 20; // Shorthand for total = total - 20
        note += "<b>Subtopic (b) Assignment:</b> <code>-=</
        code> updated the total by subtracting the discount.";
    }

    document.getElementById('outPOS').textContent = "Final
    Total: RM " + total.toFixed(2);
    showNote('notePOS', note);
}
```

4.

```
// --- TASK 3: INCREMENT & DECREMENT ---
let stock = 10;
function runTask3(type) {
    let note = "";
    if (type === 'inc') {
        stock++; // (c) Increment
        note = "<b>Subtopic (c) Increment:</b> <code>++</
        code> increased the stock by 1.";
    } else {
        stock--; // (c) Decrement
        note = "<b>Subtopic (c) Decrement:</b> <code>--</
        code> decreased the stock by 1.";
    }
    document.getElementById('stockVal').textContent = stock;
    showNote('noteStock', note);
}
```

5.

```
    // --- TASK 4: RELATIONAL & LOGICAL ---
    function runTask4() {
        let age = parseInt(document.getElementById('age').value);
        let hasParent = document.getElementById('parent').checked;
        let status = document.getElementById('outSecurity');

        if (isNaN(age)) return alert("Please enter a valid age.");

        // d. Relational: >=
        // e. Logical: || (OR), && (AND)
        let isEligible = (age >= 18) || (age >= 13 && hasParent);

        status.textContent = isEligible ? "Status: ACCESS GRANTED
        ✅" : "Status: ACCESS DENIED ❌";
        status.style.color = isEligible ? "#27ae60" : "#ff4d4d";

        let note = "<b>Subtopic (d) Relational:</b> <code>>=</
        code> compared Age (" + age + ") to the limit.<br>";
        note += "<b>Subtopic (e) Logical:</b> Used <code>||</
        code> (OR) and <code>&&</code> (AND) to combine rules.";
        showNote('noteSecurity', note);
    }
</script>
```

Capture a screenshot of the output from the previous steps. **Observe** the results and **explain** your understanding of the process.

1	**Screenshot of the output**

```
function showNote(id, text) {
    const noteDiv = document.getElementById(id);
    noteDiv.style.display = 'block';
    noteDiv.innerHTML = text;
```

	Explanation Output does not appear at it works in HTML

2.	**Screenshot of the output** **Task 1: Basic Arithmetic (Check Console)** This task demonstrates background calculations using +, -, , /, *, and %. Run Console Logic ---Task 1: Arithmetic Operations --- lab2.html:105 Final Student Count: 30 lab2.html:114 10 to the power of 2: 100 lab2.html:115 remainder of 31/2: 1 lab2.html:116
	Explanation This task demonstrates basic arithmetic operations in Javascript by calculating addition, exponentation and modulus and displaying the result in the console.

3.	**Screenshot of the output** **Task 2: POS System (Arithmetic & Assignment)** Price: RM 50.00. **Promo:** Buy > 4 units to get RM 20.00 discount. Quantity: 4 Calculate Total Final Total: RM 200.00 **Subtopic (a) Arithmetic:** * multiplied 50 by 4. Price: RM 50.00. **Promo:** Buy > 4 units to get RM 20.00 discount. Quantity: 5 Calculate Total Final Total: RM 230.00 **Subtopic (a) Arithmetic:** * multiplied 50 by 5. **Subtopic (b) Assignment:** -= updated the total by subtracting the discount.
	Explanation

	The POS system multiples the fixed price(RM50) by entered quantity and, if more than 4 units are purchased, applied a RM 20 discount using the -= assignment operator before final total.

4.	**Screenshot of the output** **Task 3: Stock Control (Increment & Decrement)** Simulating inventory updates. Stock Level: **-2** Add (+) Remove (-) **Subtopic (c) Decrement:** -- decreased the stock by 1. Simulating inventory updates. Stock Level: **3** Add (+) Remove (-) **Subtopic (c) Increment:** ++ increased the stock by 1.
	Explanation **This stock control system uses the increment(++) operator to increase the stock by 1 when "add(+)" is clicked and the decrement (--) operator to decrease the stock by 1 when "remove(-)" is clicked, updating the displayed stock level each time.**

5.	**Screenshot of the output** **Task 4: Gate Access (Relational & Logical)** Rules: Age 18+ OR (Age 13+ AND Parent present). Age: 10 ☐ With Parent? Verify Entry status ACCESS DENIED ✗ **Subtopic (d) Relational:** >= **compared Age (10) to the limit.** **Subtopic (e) Logical: Used** || **(OR) and** && **(AND) to combine rules.**

	Task 4: Gate Access (Relational & Logical) Rules: Age 18+ OR (Age 13+ AND Parent present). Age: 15 ☑ With Parent? Verify Entry status: ACCESS GRANTED ✅ **Subtopic (d) Relational: >= compared Age (15) to the limit.** **Subtopic (e) Logical: Used || (OR) and && (AND) to combine rules.** **Task 4: Gate Access (Relational & Logical)** Rules: Age 18+ OR (Age 13+ AND Parent present). Age: 7 ☑ With Parent? Verify Entry status ACCESS DENIED ❌
	Explanation **Access is allowed if the person is 18 older, or if they are at least 13 and with a parent(like age 15 with a parent) but denied if they are under 13 even with a parent (like age 7)**

3. Quick Check (Reflection)

Instruction: Answer the following questions based on your coding results.

1. **Modulus Utility:** Based on Task 1, if students % 2 returns 0, is the number even or odd?
2. **Shorthand Power:** In Task 2, rewrite total -= 20 using the standard assignment method.
3. **Logical Priority:** In Task 4, why did we put (age >= 13 && hasParent) in parentheses?
4. **Time Calculation:** (Extra Challenge) If a variable totalMinutes = 130, write the formula to find the remaining minutes after converting to hours using the **Modulus** operator.